PROJECT BRIEF / RURAL DEVELOPMENT

Problem Statement

Building a Relational Rural Self-Help Group & Microfinance Tracking Dashboard

Background

Self-Help Groups (SHGs) play an important role in rural savings, access to credit, and community development. A rural development agency must track the village clusters where groups are formed, the SHGs registered in each cluster, and the financial activity recorded for each group, including monthly savings, loan disbursements, and repayments.

Problem

The agency currently relies on fragmented spreadsheets to manage this information. As records are entered and edited by different people, the same cluster or group can be recorded inconsistently, financial entries can become detached from their owners, and staff must manually combine sheets to understand a group's activity. Deletions and corrections can also leave incomplete records or remove the wrong information. These issues make routine monitoring slow and reduce confidence in the accuracy and completeness of the data.

Project Need

The agency needs a simple, dependable tracking application that stores clusters, SHGs, and savings and loan entries as related records in a relational database. Each SHG should belong to one village cluster, and each financial entry should belong to one SHG. The system must preserve these relationships when records are removed, validate new information before saving it, and make it easy for staff to review both a combined view and the underlying tables.

Project Objective

Build a desktop dashboard using Python, Tkinter, and SQLite that replaces disconnected spreadsheet tracking with a structured, easy-to-use record system. It will support adding a group and its initial financial entry, selecting records for deletion, and reviewing current information through four database views.

Key Requirements

- Maintain unique village-cluster profiles, SHG names and formation dates, and typed financial entries with numeric amounts.
- Enforce foreign-key relationships and cascading deletion so child records cannot remain orphaned.
- Reject blank required fields and invalid or negative amounts; refresh displayed records and delete choices immediately after changes.
- Provide a joined SHG-and-ledger view and separate views of the cluster, SHG, and entry tables.

> **Expected outcome**
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> A single, traceable source of SHG and microfinance records that reduces manual reconciliation, prevents orphaned data, and helps agency staff review group activity more reliably.